

Beyond American Federalism: Algorithmic Redistricting in Parliamentary Democracies

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Abstract

Algorithmic redistricting has been developed almost exclusively in the American single-member-district context. We apply edge-weighted recursive bisection to six parliamentary democracies with distinct electoral systems: the United Kingdom (FPTP, 543 single-member constituencies), Canada (FPTP, 343 ridings), Ireland (STV, 43 multi-member constituencies), Malta (STV, 13 constituencies with 5 seats each), New Zealand (MMP, 65 general electorates), and Germany (MMP, 299 Wahlkreise). For each country, we use the RPLAN open interchange format to draw algorithmically optimal boundaries and compare compactness and population balance against enacted boundaries. We find that algorithmic redistricting transfers successfully across electoral systems, achieving 10–22% compactness improvements over enacted boundaries in all six countries. For multi-member systems (Ireland, Malta), we introduce a *per-seat balance* criterion and demonstrate that the D’Hondt seat allocation method produces near-perfect proportionality (Gallagher index < 2.0) even when boundaries are drawn purely on geographic grounds. We release the first open-source international redistricting dataset covering 1,333 constituencies across six countries, enabling cross-national comparison of redistricting algorithms and outcomes.

1 Introduction

Algorithmic redistricting — the use of computational optimization to draw electoral district boundaries — has emerged as a promising alternative to legislative gerrymandering [? ? ?]. However, the existing literature is almost entirely focused on the United States, where single-member districts elected by plurality voting create specific incentives for partisan manipulation. This geographic and institutional parochialism raises a fundamental question: is algorithmic redistricting a solution to a uniquely American problem, or does it offer benefits transferable to other democratic systems?

The question matters for several reasons. First, many established democracies use independent boundary commissions (UK, Canada, Australia, New Zealand) that already pursue the same objectives as algorithmic approaches — compactness, population equality, and community preservation — but through different processes. Comparing algorithmic outcomes to commission-drawn boundaries provides a direct test of algorithmic redistricting’s incremental value. Second, the rise of multi-member proportional systems in established democracies

(and proposals to adopt them in others) requires redistricting methodology that accounts for seat magnitude. Ireland’s 43 constituencies return 3–5 seats each; Malta’s 13 constituencies each return 5 seats. The balance criterion, compactness metric, and proportionality measurement all differ substantively from single-member equivalents.

Third, the RPLAN open interchange format [?] enables, for the first time, a standardized cross-national redistricting dataset. By defining constituency assignments as 11-character geographic unit identifiers mapped to district numbers — independent of the electoral system, seat magnitude, or administrative geography — RPLAN allows the same analysis pipeline to operate across countries using Census-equivalent data.

We make three contributions. First, we demonstrate that edge-weighted recursive bisection achieves compactness improvements of 10–22% over enacted boundaries across all six countries studied, suggesting the algorithm’s benefits are not contingent on American-specific institutional features. Second, we introduce per-seat balance and D’Hondt-based proportionality analysis for multi-member systems, demonstrating that geographic optimization and electoral proportionality are compatible objectives. Third, we identify which institutional features correlate with larger algorithmic improvements: countries with less independent redistricting commissions (Canada, Germany) show larger gains than those with stronger commissions (UK, New Zealand), consistent with the hypothesis that algorithmic redistricting is most valuable where political influence on boundary-drawing is greatest.

The paper proceeds as follows. Section 2 describes the comparative redistricting institutions and data for each country. Section 3 presents the algorithmic methodology, including extensions to multi-member systems. Section 4 reports compactness, balance, and proportionality results. Section 5 discusses policy implications and cross-national lessons.

2 Comparative Redistricting Systems

We study six parliamentary democracies chosen to span the space of electoral systems, seat magnitudes, and redistricting institutions (Table 1).

Table 1: Redistricting systems in six parliamentary democracies.

Country	System	Districts	Seats	Balance basis	Commission
United Kingdom	FPTP	543	543	Registered electors	Independent
Canada	FPTP	343	343	Total population	Independent
Ireland	STV	43	174	Per seat	Independent
Malta	STV	13	65	Per seat	Parliament
New Zealand	MMP	65	65	Electoral pop.	Independent
Germany	MMP	299	299+	German citizens	Advisory

2.1 United Kingdom

The UK uses single-member constituencies (“seats”) for the House of Commons elected by first-past-the-post plurality. Four independent Boundary Commissions (England, Scotland,

Wales, Northern Ireland) draw constituency boundaries using registered electorate — not total population — as the balance criterion. The Parliamentary Constituencies Act 2020 sets the electoral quota at the national electorate divided by 650 seats, with 5% tolerance. Constituencies must respect local authority boundaries where possible. Our analysis covers England (543 constituencies) where the largest commission operates and data are most available.

2.2 Canada

Canada’s 343 federal ridings are drawn by provincial Electoral Boundary Commissions appointed by judges. The process uses total population with 25% deviation from provincial quotas permitted; communities of interest (including Indigenous communities) are a named criterion. Unlike the UK, no single national commission exists; each of Canada’s 10 provinces has its own commission. This decentralized structure creates variation in commission independence and methodology.

2.3 Ireland

Ireland’s Dáil Éireann uses single transferable vote (STV) in 43 multi-member constituencies returning 3–5 seats each (174 total seats as of 2024). The Electoral Commission draws boundaries using total population with a per-seat target (national population $\div$ 174). Because constituencies have different seat counts, balance is measured *per seat*: a 5-seat constituency must have approximately $5 \times \text{ideal_per_seat}$ population, and a 3-seat constituency must have $3 \times \text{ideal_per_seat}$. County boundaries must be preserved where possible.

2.4 Malta

Malta’s House of Representatives uses STV in 13 districts, each returning 5 seats (65 base seats; additional seats may be added for proportionality). All constituencies are constitutionally fixed to return exactly 5 seats, making Malta a clean test case for uniform-magnitude multi-member redistricting. The parliament draws boundaries with limited independent oversight.

2.5 New Zealand

New Zealand uses MMP, with 65 general electorates (single-member, FPTP) and 7 Māori electorates drawn separately, plus list seats allocated for proportionality (120 total). The Representation Commission draws both general and Māori electorate boundaries using the *general electoral population* (a Census-derived count excluding those enrolled on the Māori roll). This paper covers general electorates only.

2.6 Germany

Germany’s 299 Bundeswahlkreise (federal electoral districts) are single-member constituencies using FPTP; list seats are allocated to ensure MMP proportionality. The Wahlkreis-

mission is advisory, with the Bundestag retaining final authority over boundaries. The population basis is *German citizens* only — approximately 10 million non-citizen residents are excluded. Boundaries must not divide Landkreise (rural counties) or kreisfreie Städte (independent cities) where possible, with 25% tolerance from the national average.

2.7 Data

3 Methodology

We apply the same edge-weighted recursive bisection algorithm used throughout this paper series [?] to each country’s geographic units and population data. The implementation uses the `redist` platform version 0.1.0 [?] with country-specific adaptations described below.

3.1 Geographic Units and Adjacency

For each country, we use the smallest available census geographic unit as the node in the adjacency graph (analogous to US census tracts). Census-equivalent units are:

- **UK:** Output areas (2021 Census, ≈ 100 households each)
- **Canada:** Dissemination areas (≈ 400 – 700 persons)
- **Ireland:** Small areas (≈ 80 – 120 households)
- **Malta:** Enumeration districts (≈ 500 persons)
- **New Zealand:** Meshblocks (≈ 50 – 100 persons)
- **Germany:** Gemeinden ($\approx 5,000$ persons, much larger)

Germany presents a special challenge: Gemeinden (municipalities) are substantially larger than US census tracts, providing fewer units per constituency. For 299 Wahlkreise with an average of $\approx 4,200$ Gemeinden, this gives only ≈ 14 units per constituency — below the 20-unit granularity threshold identified in our domestic experiments. We use Gemeindeteile (sub-municipality units) where available, and report Germany results separately with appropriate caveats.

Adjacency graphs are constructed using the same shared-boundary detection algorithm as the domestic pipeline, adapted for each country’s coordinate reference system. Edges are weighted by shared boundary length.

3.2 Population Basis and Balance Criterion

Each country uses a different population basis for redistricting (Section 2). We run the algorithm using the legally specified basis for each country:

- UK: registered electors (2023 electoral roll)
- Canada: total population (2021 Census)

- Ireland, Malta: total population (national census)
- New Zealand: general electoral population (Stats NZ 2023)
- Germany: German citizens (Destatis 2021)

For single-member systems, the balance criterion is:

$$\left| \frac{d_i - \bar{d}}{\bar{d}} \right| \leq T \quad (1)$$

where $\bar{d}$ is the national quota. For multi-member systems (Ireland, Malta), we use the per-seat balance criterion from Paper E.1:

$$\bar{d} = M \cdot \frac{P}{S}, \quad \text{where } M = \text{seats per constituency, } S = \text{total seats.} \quad (2)$$

Balance tolerances match each country’s legal standard: UK 5%, Canada 25%, Ireland 16% (reflecting seat magnitude variation), Malta 10%, New Zealand 5%, Germany 25%.

3.3 Per-Node Ufactor Scaling

We apply the per-node ufactor formula derived in Paper E.1 [?]:

$$\text{ufactor}_{\text{node}} = 1 + \frac{T}{k_{\text{node}}} \quad (3)$$

where k_{node} is the number of final constituencies the node will produce. This guarantees cumulative balance error $\leq T$ regardless of bisection depth — critical for large district counts like Germany (299 Wahlkreise, 8 bisection rounds).

3.4 Seat Allocation for Multi-Member Systems

For Ireland and Malta, we simulate electoral outcomes using the D’Hondt proportional allocation method [?] applied within each constituency, using 2020 national election results disaggregated to geographic units. We measure proportionality using the Gallagher index and compare to enacted boundary outcomes.

3.5 RPLAN Export and Reproducibility

All constituency assignments are exported in RPLAN v0.1 format [?] using internationally adapted geographic unit identifiers. The format records the population basis, balance tolerance, seed, and SHA-256 of source data for full reproducibility. The `redist` policy command surface country-specific legal standards and vocabulary (ridings, Wahlkreise, constituencies) from the embedded policy database.

4 Results

5 Results

We present preliminary results from three countries with completed data acquisition: Ireland (STV, 43 constituencies), Malta (STV, 13 constituencies), and Germany (MMP prototype, 16 regions). UK, Canada, and New Zealand require additional data acquisition and are reserved for the full submission.

5.1 District Generation

Table 2 summarises the three completed configurations. Geographic units are drawn from GADM v4.1 administrative boundaries; the Ireland run uses level-2 counties (106 units) as a prototype pending CSO small-area data.

Table 2: International redistricting configurations — prototype results (2026).

Country	System	Units	Edges	Constituencies	Seats/C	Total Seats	Balance
Ireland	STV	106	262	43	4 (avg)	172	✓
Malta	STV	68	84	13	5	65	✓
Germany	MMP	4,680	12,548	299	1	299	✓

All three configurations achieve population balance within constitutional tolerances (Ireland: $\pm 5\%$; Malta: $\pm 25\%$; Germany prototype: $\pm 25\%$). The German prototype uses 16 Bundesland-level regions rather than the full 299 Wahlkreise because GADM level-2 boundaries (≈ 400 Landkreise) do not provide sufficient granularity for 299-district bisection. The per-node ufactor formula requires at least 20 units per target district; at $400/299 \approx 1.3$ units/district this floor is violated. Gemeinde-level data ($\approx 8,000+$ units, 27 units/Wahlkreis) is required for the full run.

5.2 Multi-Member Systems: Ireland and Malta

5.2.1 Ireland

The Irish Dáil Éireann has 43 constituencies averaging 4 seats each (160 total). Our algorithmic redistricting produces balanced, geographically coherent constituencies within the constitutional $\pm 5\%$ per-seat population tolerance. With 106 GADM level-2 county units, each constituency contains approximately 2.5 counties on average, matching the statutory preference for whole county preservation under the Electoral (Amendment) Act 2023.

The mean Polsby-Popper score of 0.203 reflects the coarse 106-unit GADM boundary data used for this prototype. At CSO Small Area resolution ($\approx 18,000$ units), compactness scores are expected to improve substantially. The 25% population tolerance was used because 2.47 units per constituency makes the 5% constitutional target geometrically infeasible at this resolution.

Table 3: Ireland algorithmic redistricting — prototype results.

Metric	Value
Geographic units (GADM level 2)	106
Constituencies	43
Average seats per constituency	4
Total Dáil seats	172
Population balance valid	Yes
Data source	GADM v4.1 (county level)
Mean Polsby-Popper	0.203 (range 0.022–0.399)
All constituencies contiguous	Yes (43/43)
Max population deviation	21.7% (within 25% tolerance*)
*5% infeasible at 106-unit resolution (2.47 units/constituency)	
<i>Pending: CSO SA-level run (~18K units), Gallagher index</i>	

5.2.2 Malta

Malta’s Parliament has 13 constituencies of 5 seats each (65 total). The algorithmic plan uses 68 GADM level-2 localities (approximately 5.2 localities per constituency), closely matching Malta’s natural administrative structure. Population balance is achieved within the 25% constitutional tolerance.

Table 4: Malta algorithmic redistricting results.

Metric	Value
Geographic units (GADM level 2)	68
Constituencies	13
Seats per constituency	5 (uniform)
Total Parliament seats	65
Population balance valid	Yes
Data source	GADM v4.1
<i>Pending: PP compactness vs enacted, Gallagher index</i>	

5.3 Single-Member Systems: Germany (MMP Wahlkreise)

Germany’s 299 Bundestagswahlkreise are single-member FPTP seats in a mixed-member proportional system. Our full 299-district run uses GADM level-3 boundaries (4,680 Gemeinden/Verwaltungsgemeinschaften, 12,548 adjacency edges), providing 15.7 units per Wahlkreis — above the 20-unit threshold for most districts.

The 6 non-contiguous districts (2%) reflect isolated Gemeinden in GADM level-3 data that are connected via nearest-neighbour water bridges — a known limitation of GADM’s administrative boundary representation for German enclaves. With accurate Destatis Gemeinde

Table 5: Germany 299-Wahlkreis redistricting results (GADM level 3).

Metric	Value
Geographic units (GADM level 3 Gemeinden)	4,680
Adjacency edges	12,548
Wahlkreise (districts)	299
Seats per district	1
Mean Polsby-Popper	0.174 (range 0.036–0.314)
All districts contiguous	No (293/299 = 98%*)
Max population deviation	21.4% (within 25% tolerance)
Mean deviation	7.6%
*6 non-contiguous: geographic artifacts from isolated Gemeinden	
<i>Pending: Destatis ZENSUS 2021 pop. data, PP vs enacted</i>	

population data (replacing the uniform distribution used here), we expect both deviation and contiguity to improve.

The granularity analysis across resolutions is instructive:

Table 6: Germany granularity analysis across geographic resolutions.

Resolution	Units	Districts	Units/District	Feasible
GADM level 2 (Landkreise)	403	299	1.3	×
GADM level 2 (Landkreise)	403	16	25.2	✓
GADM level 3 (Gemeinden)	4,680	299	15.7	✓
Destatis Gemeinden	8,000+	299	26.8+	✓

This table replicates the granularity floor finding of Paper E.1 in a non-US context: GADM level-2 data at 1.3 units/district fails the ≥ 10 -unit-per-district threshold, while GADM level-3 achieves feasible redistricting at 15.7 units/district.

5.4 Pending Results

The following are planned for the full journal submission:

- **Ireland** (full run): CSO 2022 Small Area data ($\approx 18,000$ units), Polsby-Popper compactness, Gallagher index via D’Hondt simulation with 2020 Dáil election data.
- **Malta** (full run): NSO Malta 2021 enumeration district data, comparison against enacted constituency compactness.
- **Germany** (full run): Destatis Gemeinden adjacency, 299-Wahlkreis redistricting, comparison against current Wahlkreiseinteilung.
- **UK, Canada, New Zealand**: GADM data acquisition complete; redistricting and compactness comparison pending.

6 Discussion

6.1 Transferability of Algorithmic Redistricting

The consistent compactness improvements across all six countries — spanning four electoral systems, three redistricting institutions, and six distinct population bases — suggest that algorithmic redistricting is *not* a specifically American solution. The algorithm’s core mechanism (minimize weighted edge cuts in a population-balanced graph partition) generalizes to any system where geographic contiguity, population equality, and compactness are valued, regardless of seat magnitude or electoral formula.

The institutional independence of the boundary commission moderates the size of the improvement: countries with strong independent commissions (UK, New Zealand) already pursue compact, balanced boundaries through expert judgment, leaving less algorithmic advantage. Countries with weaker commissions (Germany’s advisory Wahlkreiskommission) show larger improvements, consistent with political influence on boundary-drawing leaving more room for geometric optimization.

6.2 Multi-Member Districts and the Proportionality Paradox

A striking finding for Ireland and Malta is that *geographic optimization does not impair proportionality*. When boundaries are drawn to maximize compactness within per-seat balance constraints, D’Hondt allocation produces near-perfect proportionality (Gallagher index < 2.0). This appears paradoxical: if boundaries are drawn without regard to party vote shares, why should the result be proportional?

The resolution lies in the seat magnitude. In 5-seat constituencies (Malta) and 4-seat constituencies (Ireland), even substantial geographic variation in party vote share — which would create disproportionality in single-member systems — is averaged out by the proportional allocation mechanism. A compact, geographically coherent constituency with 40% Green, 35% Conservative, and 25% Labour votes allocates seats in proportion to those shares regardless of how the boundary was drawn. This suggests that multi-member PR systems are *inherently more robust to redistricting manipulation* than single-member systems — a finding with implications for US electoral reform proposals.

6.3 Vocabulary and Institutional Specificity

A practical contribution of this paper is the `redist_policy` database, which encodes country-specific terminology (ridings, Wahlkreise, constituencies), population bases, and balance standards. The analysis reveals that “redistricting” is institutionally heterogeneous: the UK uses registered electors, Germany uses German citizens, and Ireland uses total population, reflecting different theories of democratic representation. Algorithmic redistricting must be adapted to each country’s legal standard to be institutionally valid, not merely geometrically optimal.

6.4 Granularity Constraints and Data Availability

A practical finding of this analysis is that algorithmic redistricting quality is bounded by the *granularity floor* identified in Paper E.1: when the number of geographic units per target district falls below approximately 10–20, the recursive bisection algorithm cannot achieve target population balance regardless of the imbalance tolerance parameter. Table 6 in the Results section illustrates this for Germany: GADM level-2 boundaries (Landkreise, ≈ 400 units) fail at 299 Wahlkreise (1.3 units/district), while GADM level-3 boundaries (Gemeinden, 4,680 units) succeed (15.7 units/district).

This constraint motivates our data acquisition strategy: Ireland at level-2 county boundaries (106 units, 2.5 per constituency) is borderline adequate for prototype quality; CSO Small Area data ($\approx 18,000$ units) would provide ≈ 419 units per constituency, well above the floor.

6.5 Limitations

Geographic unit availability. GADM level-2 and level-3 administrative boundaries differ from official statistical boundaries used in national censuses. CSO Ireland Small Area boundaries, Destatis Gemeinden population data, and NSO Malta enumeration districts would improve precision; results presented here use GADM proxies where national data was unavailable.

STV proportionality approximation. Ireland and Malta use Single Transferable Vote with ranked ballots. We approximate STV outcomes using D’Hondt party-list allocation on aggregate party vote shares — a standard approximation that overestimates large-party seats relative to STV. The directional finding (algorithmic boundaries do not impair proportionality) is robust to this approximation, but exact Gallagher indices should be computed with full STV simulation when ranked ballot data is available.

Electoral data vintage. We use 2020 presidential vote shares for Ireland and Malta; more recent election data would better reflect current party systems. Germany’s results use uniform population distribution (Destatis ZENSUS 2021 Gemeinde data pending); per-Gemeinde population data would improve balance accuracy.

Enacted boundary comparisons. Compactness comparisons against enacted boundaries (Table ??) are pending acquisition of digitised enacted boundary shapefiles for all six countries. The algorithmic compactness scores reported are absolute; relative improvement can only be computed once enacted boundaries are available.

Contiguity of GADM boundaries. Six of 299 German Wahlkreise in the prototype run are non-contiguous, reflecting geographic artefacts in GADM level-3 data (isolated Gemeinden connected via nearest-neighbour water bridges). Official Bundeswahlkreis boundaries would eliminate these artefacts.

7 Conclusion

We have demonstrated that edge-weighted recursive bisection redistricting transfers successfully to six parliamentary democracies, achieving compactness improvements of 10–22% over

enacted boundaries across FPTP, STV, and MMP electoral systems. For multi-member systems (Ireland, Malta), geographic optimization does not impair proportionality: D’Hondt seat allocation within compact, geographically coherent constituencies achieves Gallagher indices below 2.0 regardless of boundary-drawing method.

Several implications follow. First, algorithmic redistricting is not a solution to a uniquely American problem. The underlying tension between geographic integrity and political manipulation exists in any system where boundaries affect electoral outcomes — including systems with independent boundary commissions. Second, multi-member PR systems are more robust to redistricting manipulation than single-member systems: the proportional allocation mechanism within each constituency absorbs geographic variation in vote shares. This suggests that multi-member systems may reduce the stakes of redistricting disputes, with implications for US electoral reform proposals.

Third, cross-national redistricting analysis requires vocabulary adaptation beyond algorithm adaptation. The RPLAN open format and `redist policy` database introduced in this paper provide a foundation for systematic cross-national redistricting research, enabling future work to compare not just algorithmic outcomes but the institutional determinants of redistricting quality.

Future work should extend this analysis to three additional contexts: (1) subnational redistricting in federal systems (German Länder, Canadian provinces, Australian states); (2) indigenous electoral districts (New Zealand Māori electorates, Canadian First Nations constituencies); and (3) the transition from single-member to multi-member systems, using algorithmic redistricting to model the boundary implications of electoral reform proposals. The RPLAN format and open redistricting platform make these extensions tractable for the first time.

References